

Voice Transcription and Identification for Tactical Radio Networks

Mission Analysis

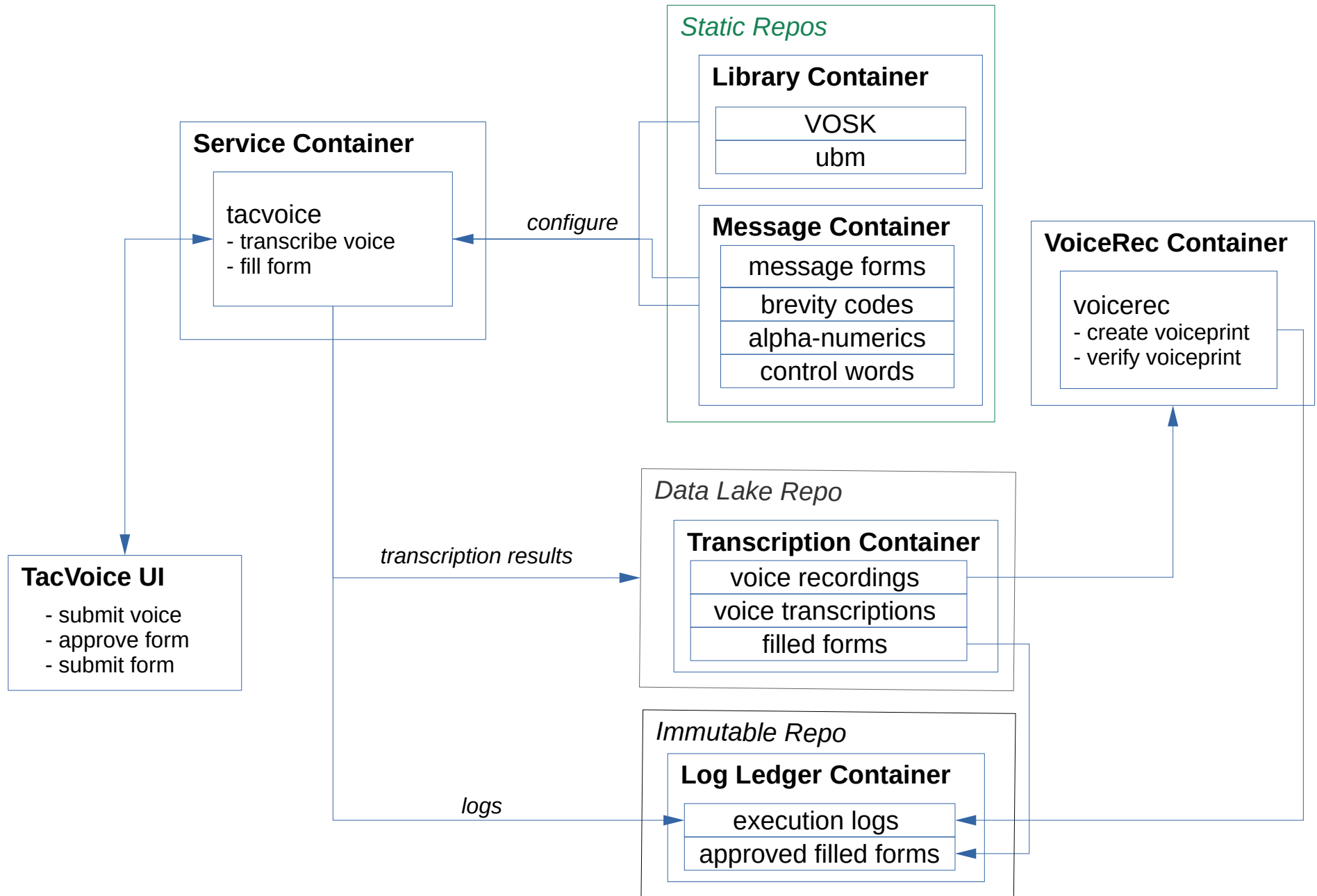
1. Voice Transcription is achieved using inference libraries and base models for speech recognition. Accuracy is further improved by providing a list of words to expect. Each of these must be controlled in an operational environment.
2. If recordings are to be used for learning or improvement, then this must be handled by an independent application that will extract voice and transcription data from the temporary “Data Lake” storage service. In this case retention and handling will be determined by an external policy, and access controls will be assigned independently.
3. Even if voice recordings are not used after transcription, it is important to specify how they are handled so that policy requirements associated with retention of biometric data can be verified.
4. The use of containers to separated concerns provides a way to establish a secure local capability that can be accessed by services in external clusters using policy configurations, and selectively improved without re-creating the core services.

Execution

1. Plan a secure mTLS network for a network of (6) containers for separation of Library, Voice, Message, Transcription, and Storage concerns, (1) container for the TacVoice transcription service and (1) container for the voice recognition/verification service.
2. Create a Policy that specifies access controls for all concerns and assigns keys to all nodes.
3. Deploy all services using an HHQ provisioned cluster.
4. Expose user interface functions using WASM that supports web or mobile integration.

TacVoice: Voice Transcription and Identification for Tactical Radio Networks

Container Architecture



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Security Policy Requirements

Information about Operators – to include image, voice, and other biometrics falls under Force Protection rules which far exceed the highest classification concerns at the operational level.

Unit Level Policy

1. Voice Transcription Libraries and operational configuration information must not be retained on disk after they are provisioned from Static Repositories.
2. Voice recordings must not be retained on disk.
3. Voice transcriptions must not be retained on disk.
4. Logs will include time, mission, message name and approval status.
6. Approved messages and logs will be retained using an external service.
7. Once a specified mission period is over, all Pods will be destroyed.

Higher Headquarters Policy

1. All unit transcription services will use the voice transcription libraries and operational configuration information provisioned by HHQ.
2. Voice recordings and transcriptions submitted to the Data Lake may be assessed by HHQ services for purposes of voice recognition and model improvement.
3. Approved messages and logs will be submitted for retention by HHQ services.
7. No local storage of voice recordings or form content is permitted.

Voice Transcription and Identification for Tactical Radio Networks Systems View

